



Matching, turn C into Cx, then cancel

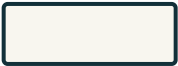








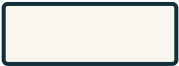




RZ0010



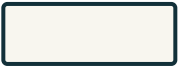




R2021



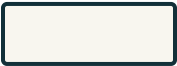




R2023)







R204





RZ/O5)

R206)

$$M = \begin{bmatrix} p & ip & 0 & 0 \\ p & -ip & 0 & 0 \\ 0 & 0 & r & ir \\ 0 & 0 & r & -ir \end{bmatrix}$$

$$\alpha = \sum \theta_i$$

$$p = e^{(-i\alpha/2)} / \sqrt{2}$$

Synthesized S

$$\begin{bmatrix} a & b & c & 0 \\ d & e & f & 0 \\ g & h & j & k \\ g & h & j & -k \end{bmatrix}$$

$$r = e^{(i\alpha/2)} / \sqrt{2}$$

$$SX^5 = SX$$

$M \models S?$







CX



$$\mathbf{M} = \mathbf{S}[\mathbf{a}, \dots, \mathbf{k}]$$

$$\mathbf{a} = \mathbf{d} = \mathbf{p}$$

$$\mathbf{b} = i\mathbf{p}, \quad \mathbf{e} = -i\mathbf{p}$$

$$\mathbf{j} = \mathbf{r}, \quad \mathbf{k} = i\mathbf{r}$$

$$\mathbf{c} = \mathbf{f} = \mathbf{g} = \mathbf{h} = 0$$

CZ



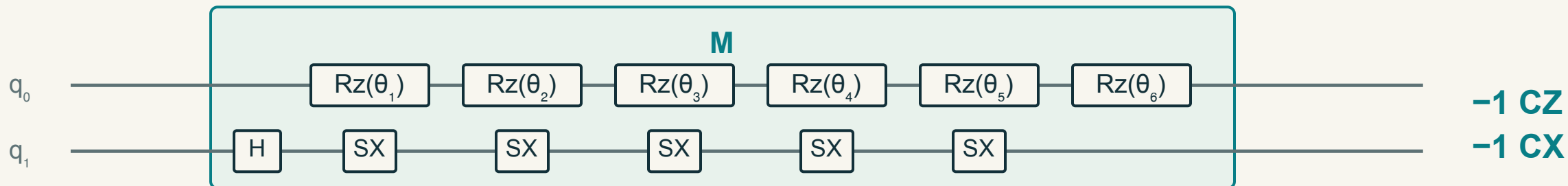
CX



-1 CZ

-1 CX

Match a long S, turn CZ into CX, then cancel



$$M = \begin{bmatrix} p & ip & 0 & 0 \\ p & -ip & 0 & 0 \\ 0 & 0 & r & ir \\ 0 & 0 & r & -ir \end{bmatrix} \xrightarrow{M \models S?} \begin{bmatrix} a & b & c & 0 \\ d & e & f & 0 \\ g & h & j & k \\ g & h & j & -k \end{bmatrix}$$

$\alpha = \sum \theta_i$
 $p = e^{(-i\alpha/2)} / \sqrt{2}$
 $r = e^{(i\alpha/2)} / \sqrt{2}$

Synthesized S

$M = S[a, \dots, k]$
 $a = d = p$
 $b = ip, e = -ip$
 $j = r, k = ir$
 $c = f = g = h = 0$

$SX^5 = SX$

Anchoring exposes the adjacent pair

Canonical rule

